Lesson 1.4 - Statistic vs Parameter and Population vs Sample

> **Video:** Statistic vs Parameter & Population vs Sample by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

Population vs sample

- **Population:** *everyone or everything* in the group being studied.
- **Sample:** a **subset**, a smaller part of the population that is actually measured.

Statistic vs parameter

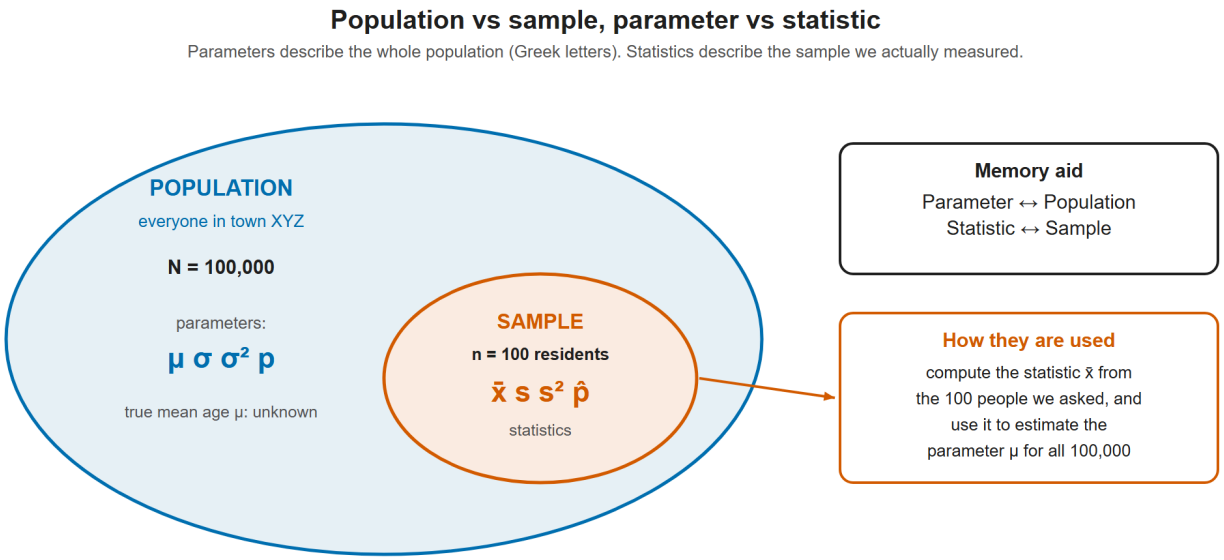
- **Parameter:** a number that describes the **population**.
- **Statistic:** a number that describes a **sample**.

> **Memory aid:** **P**arameter ↔ **P**opulation, **S**tatistic ↔ **S**ample.

Example - average age in town XYZ

Town XYZ has **100,000** residents, and we want their average age.

- Asking all 100,000 people is a lot of work, so we ask **100** residents and average their ages.
- The average age of those 100 people is the **sample mean** $\bar{x}$, which is a **statistic**.
- The true average age of all 100,000 residents is the **population mean** μ, which is a **parameter**.
- Population size = 100,000; sample size = 100.



The sample is part of the population. Numbers computed from the sample (xÌ□, s, pÌ□) are statistics, and we use them to estimate the population's parameters (Î¼, Ï□, p).

Symbols

Quantity	Sample (statistic)	Population (parameter)
Mean	$\bar{x}$ ("x-bar")	μ ("mu")
Standard deviation	s	σ ("sigma")
Variance	s^2	σ^2
Proportion	$\hat{p}$ ("p-hat")	p
Size	n (lowercase)	N (uppercase)

Roman letters ($\bar{x}$, s, $\hat{p}$, n) are usually used for samples. Greek letters (μ, σ) and plain p, N are used for populations.

Quick quiz - statistic or parameter?

A parameter describes **everyone** in the group. A statistic describes **only some** of them.

#	Statement	Answer	Symbol
1	The average weight of **all** males in the United States	**Parameter**	μ
2	The average height of **100 cats** in California (out of millions)	**Statistic**	$\bar{x}$

#	Statement	Answer	Symbol
3	The average test score of **20 students** in a class of 500	**Statistic**	$\bar{x}$
4	The average test score of **all** students in the class	**Parameter**	μ

Key takeaways

- **Population** = the whole group. **Sample** = the part you actually observe.
- **Parameter** describes a population. **Statistic** describes a sample.
- In practice we usually compute statistics ($\bar{x}$, s, $\hat{p}$) to **estimate** unknown parameters (μ, σ, p).